

CS258 Computer Communication Systems- TCP Sliding Window

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Output Screenshots:

Server:

The screenshot shows:

- 1) Number of Packets Received
- 2) Server IP Address
- 3) Average Good Put



The screenshot shows a terminal window with a dark background and light-colored text. At the top, there are four tabs: 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', and 'TERMINAL'. The 'TERMINAL' tab is selected and underlined. The terminal output consists of 20 lines of ' [>] Received [number] packets...' where the numbers range from 9933700 to 9994300 in increments of 1000. After these lines, the text 'Execution completed successfully' appears, followed by 'Server IP: 192.168.1.71', 'Total packets received: 10000000', 'Average goodput: 0.9474', and 'Total runtime: 97.26 seconds'.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

[>] Received 9933700 packets...
[>] Received 9934600 packets...
[>] Received 9938400 packets...
[>] Received 9940500 packets...
[>] Received 9952600 packets...
[>] Received 9953300 packets...
[>] Received 9953500 packets...
[>] Received 9955500 packets...
[>] Received 9956500 packets...
[>] Received 9964300 packets...
[>] Received 9965600 packets...
[>] Received 9966700 packets...
[>] Received 9969600 packets...
[>] Received 9972400 packets...
[>] Received 9978500 packets...
[>] Received 9986100 packets...
[>] Received 9986300 packets...
[>] Received 9986500 packets...
[>] Received 9987200 packets...
[>] Received 9990900 packets...
[>] Received 9991700 packets...
[>] Received 9992200 packets...
[>] Received 9994300 packets...
Execution completed successfully
Server IP: 192.168.1.71
Total packets received: 10000000
Average goodput: 0.9474
Total runtime: 97.26 seconds
```

Client:

The screenshot shows:

- 1) Number of Packets Sent
- 2) Client IP Address
- 3) Number of Retransmissions

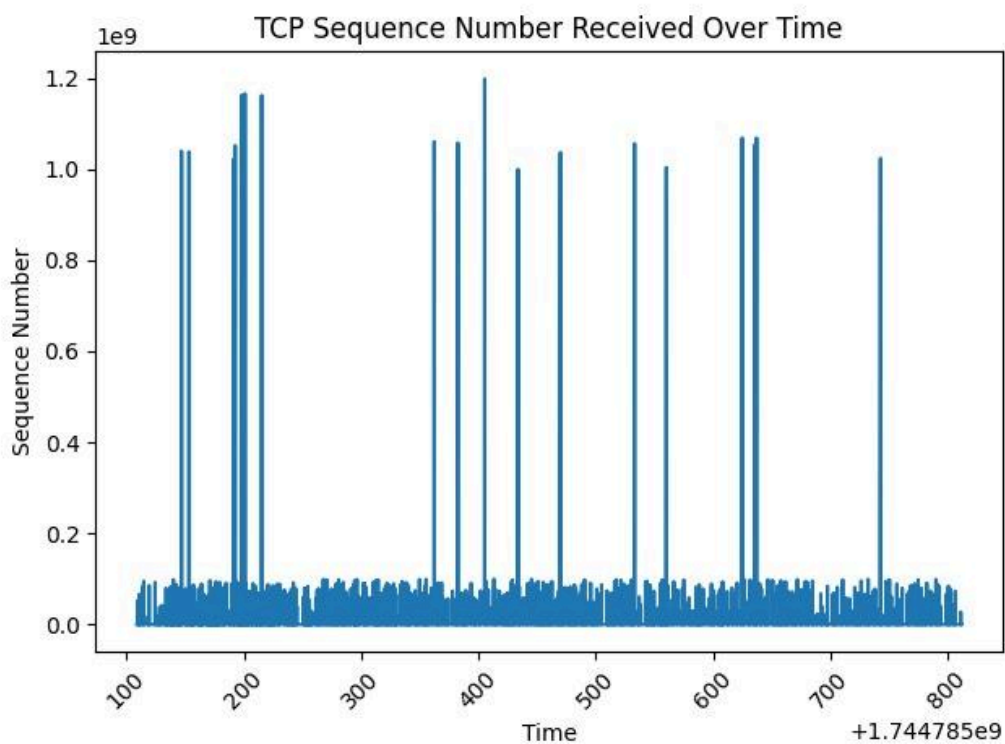
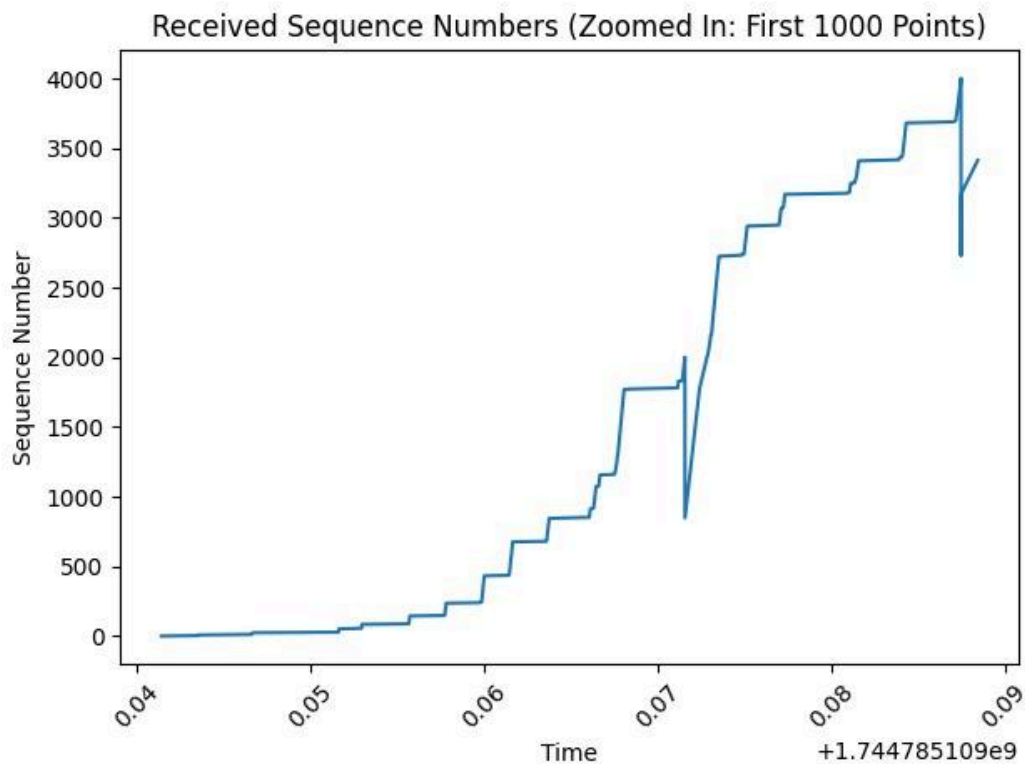
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

[>] Progress: 10054840 packets sent, Window size: 295
[>] Progress: 10056900 packets sent, Window size: 525
[>] Progress: 10058380 packets sent, Window size: 247
[>] Progress: 10060650 packets sent, Window size: 370
[>] Progress: 10061020 packets sent, Window size: 413
[>] Progress: 10062300 packets sent, Window size: 386
[>] Progress: 10064340 packets sent, Window size: 341
[>] Progress: 10067860 packets sent, Window size: 515
[>] Progress: 10069130 packets sent, Window size: 301
[>] Progress: 10071670 packets sent, Window size: 333
[>] Progress: 10072920 packets sent, Window size: 217
[>] Progress: 10080110 packets sent, Window size: 385
[>] Progress: 10083920 packets sent, Window size: 299
[>] Progress: 10084660 packets sent, Window size: 266
[>] Progress: 10088460 packets sent, Window size: 214
[>] Progress: 10090220 packets sent, Window size: 670
[>] Progress: 10090890 packets sent, Window size: 707
[>] Progress: 10097950 packets sent, Window size: 562
[>] Progress: 10100190 packets sent, Window size: 749
Execution completed successfully
Client IP: 192.168.1.179
Total packets sent: 10100966
Retransmission attempts by count:
  1 attempt: 99984
  2 attempts: 967
  3 attempts: 15
  4+ attempts: 0
Total runtime: 98.28 seconds
```

Graphs:

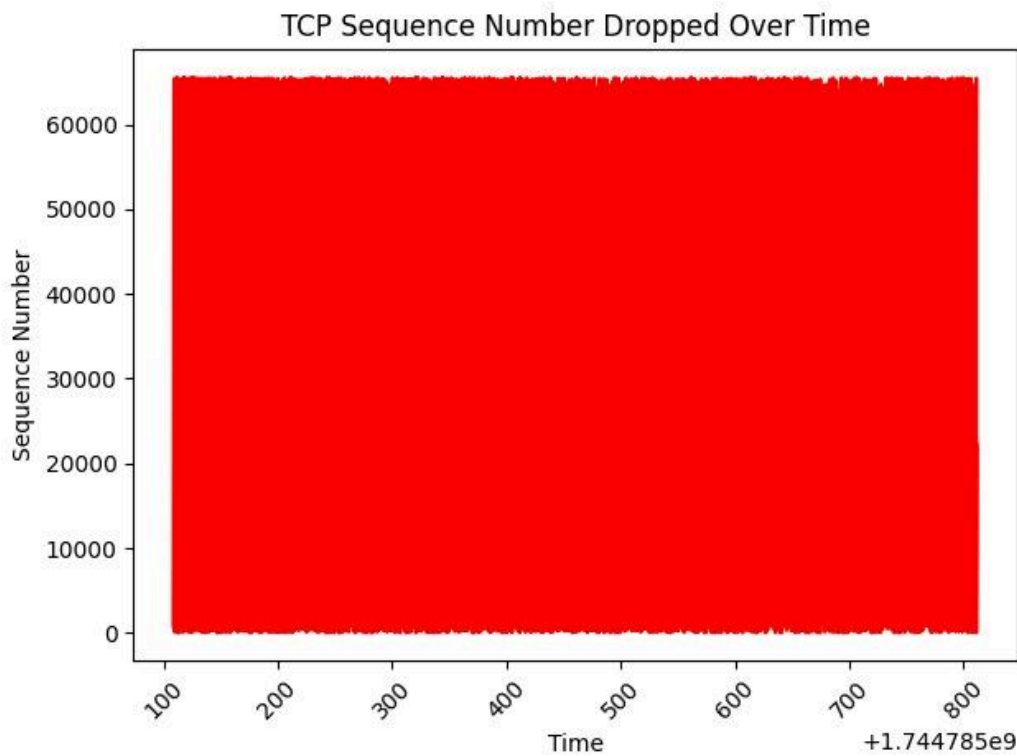
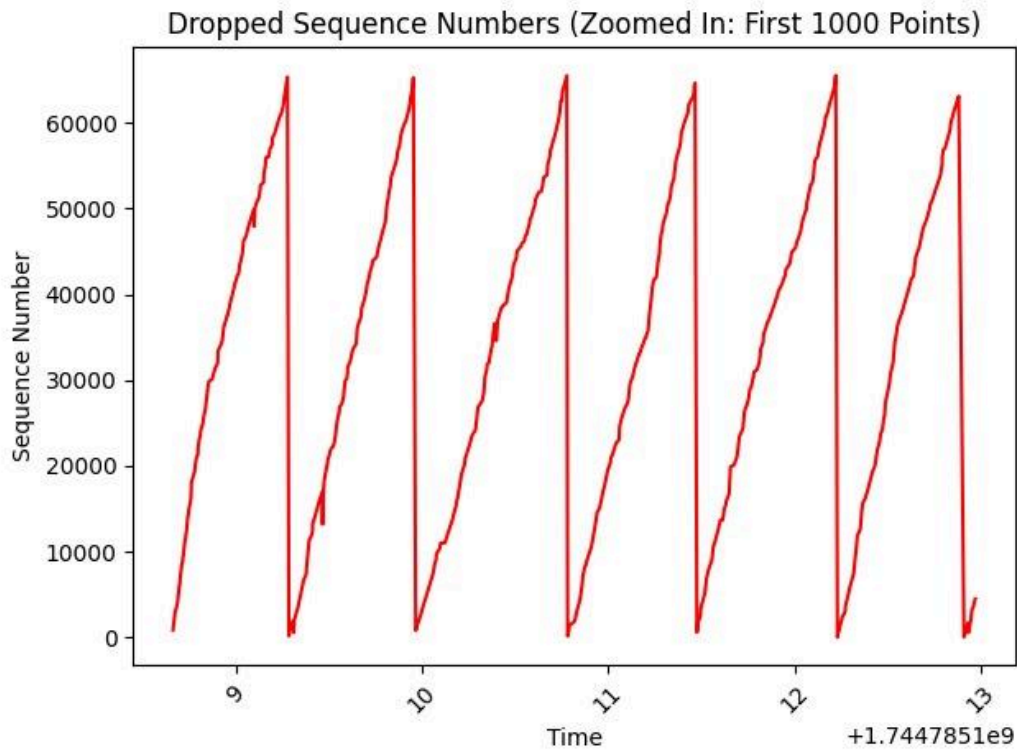
Sequence number received against time:

Note: These graphs display a number of packets in ascending order, with Graph 1 representing a small fraction of the data and Graph 2 showing all the data.



Dropped packets over time:

Note: These graphs display a number of packets in ascending order, with Graph 1 representing a small fraction of the data and Graph 2 showing all the data.



Window size change over time :

Note: These graphs display a range of time in ascending order, with Graph 1 representing a short time period and Graph 2 showing the entire duration.

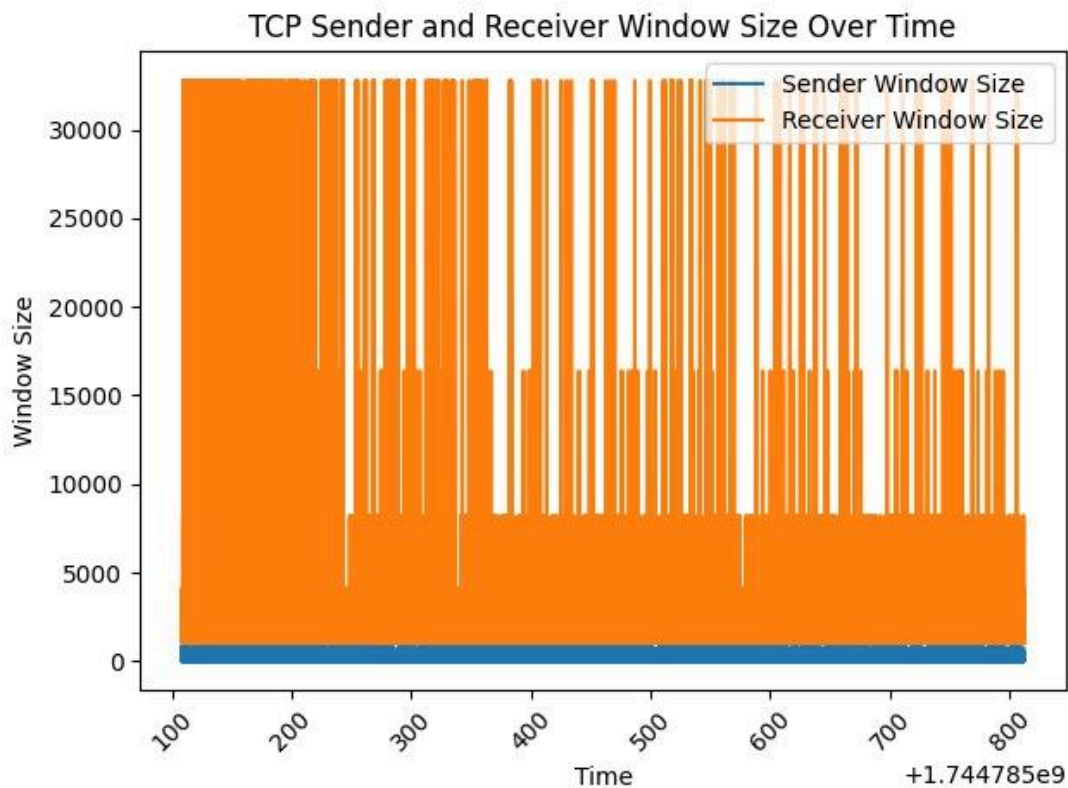
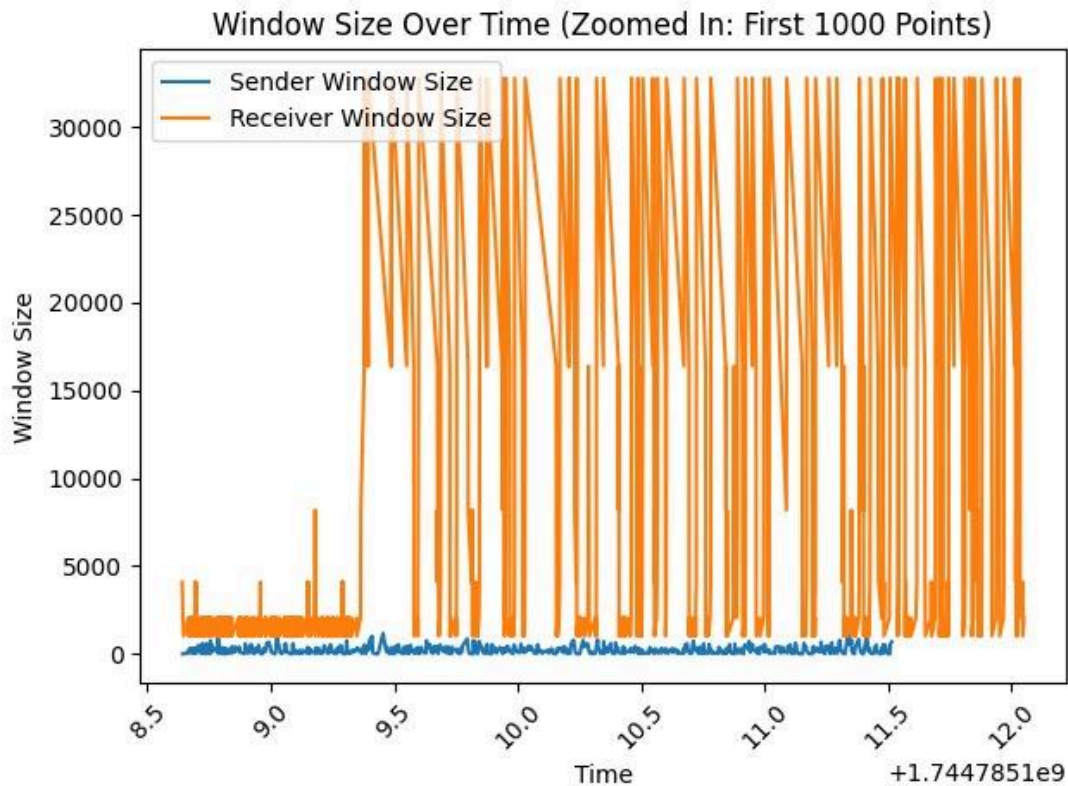


Table for retransmissions:

# of retransmissions	# of packets
1	99984
2	967
3	15
4	0

This screenshot is a part of the client output, showing the results in the table: